

AI Plant Disease Detection System - Project Roadmap

Current Status

Model trained on 15 classes with ~94-95% validation accuracy. Flask web app completed with image upload, disease prediction, confidence score, descriptions, and treatments.

Phase 1 - Immediate Improvements

1. Add Top-3 Predictions
2. Improve confidence filtering (85-90%)
3. Better UI/UX and disease cards
4. Store prediction history

Phase 2 - Computer Vision Enhancements

1. Grad-CAM heatmaps
2. Leaf detection using YOLOv8
3. Automatic leaf cropping
4. Background removal

Phase 3 - Dataset Improvements

1. Add real-world farm images
2. Collect images from Google, Kaggle, farmers
3. Increase dataset to 50k+ images
4. Apply augmentation

Phase 4 - Smart Agriculture Features

1. Disease severity estimation (Low/Medium/High)
2. Fertilizer recommendations
3. Treatment recommendations
4. Weather-based disease alerts

Phase 5 - Multi-Crop Support

Add Apple, Corn, Grape, Rice, Wheat, Cotton, Mango, etc.

Phase 6 - Deployment

1. Deploy on Render/Railway

2. Create REST API
3. Mobile-friendly interface
4. Build Android app

Final Goal

Transform the project from a simple PlantVillage classifier into a production-grade AI Agriculture Assistant.